Strategic Resource Optimization: Chicago Crime Analysis

Course: DSA 1080B – Programming for Data Science

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1. Executive Summary

This project analyzes the 2026 Chicago Crime dataset to identify patterns of criminal activity and police response efficiency. This analysis focuses on **resource optimization**. By identifying 'efficiency gaps' in arrest rates across different times and locations, a data driven model is developed for distributing existing resources to maximize the safety of the city's residents without increasing the city's budget

2. Introduction

2.1. Project Purpose

The purpose of this analysis is to evaluate the spatial and temporal distribution of criminal activity in Chicago during early 2026 to identify opportunities for **resource optimization**. While traditional crime reporting focuses on volume, this study prioritizes **arrest efficiency**—the ratio of arrests to total incidents—to determine where and when the Chicago Police Department (CPD) can achieve the "maximum positive effect" with its current force

2.2. Problem Statement

Public safety departments often face the challenge of static resource allocation in a dynamic environment. In Chicago, a "one-size-fits-all" deployment model fails to account for the drastic differences in crime types across various locations and the sharp fluctuations in incident frequency throughout the day. The primary problem addressed in this report is the **efficiency gap** identified in our initial data exploration: high-volume districts often suffer from lower arrest rates, suggesting that resources are saturated and incorrectly distributed

2.3. Scope and Objectives

The scope of this analysis is limited to the first quarter of 2026, utilizing a dataset of over 36,000 recorded incidents. The specific objectives are:

- **Identify Critical Windows:** Pinpoint the exact hours where crime frequency is highest to suggest dynamic shift adjustments.

- **Map Resource Saturation:** Use district ranking to identify areas where the arrest-to-crime ratio is significantly below the city average.
- **Segment by Location:** Categorize incidents by "Location Description" to differentiate between public deterrence needs (Streets/Sidewalks) and investigative response needs (Residences).
- **Deliver Actionable Reallocation Strategies:** Propose three resource-neutral recommendations that maximize public safety outcomes using existing data.

3. Methodology

3.1. Data Source and Description

The primary dataset for this analysis was sourced from the **City of Chicago Data Portal**, specifically the "Crimes - 2026" repository.

- **Scope:** The data consists of over **36,000 recorded incidents** spanning the first quarter of 2026.
- **Granularity:** Each record contains 22 variables, including the unique incident ID, primary crime type, arrest status, domestic flag, and precise temporal/spatial markers (Date, Hour, District, Latitude/Longitude).

3.2. Data Cleaning and Preprocessing

To ensure the integrity of our resource optimization model, the raw dataset underwent a rigorous cleaning process using the Python **Pandas** library:

- **Temporal Standardization:** The Date string column was converted into a proper datetime format. This allowed for the extraction of the Hour and Month features, which are critical for temporal demand analysis.
- **Missing Value Management:** Identified missing values in the Location Description field and categorized them as "UNKNOWN" to prevent data loss. Additionally, rows with missing geographical coordinates were removed to ensure the accuracy of spatial rankings.
- **Integrity Checks:** Performed a deduplication check based on the unique ID column, ensuring that no single crime incident was double-counted in our efficiency metrics.

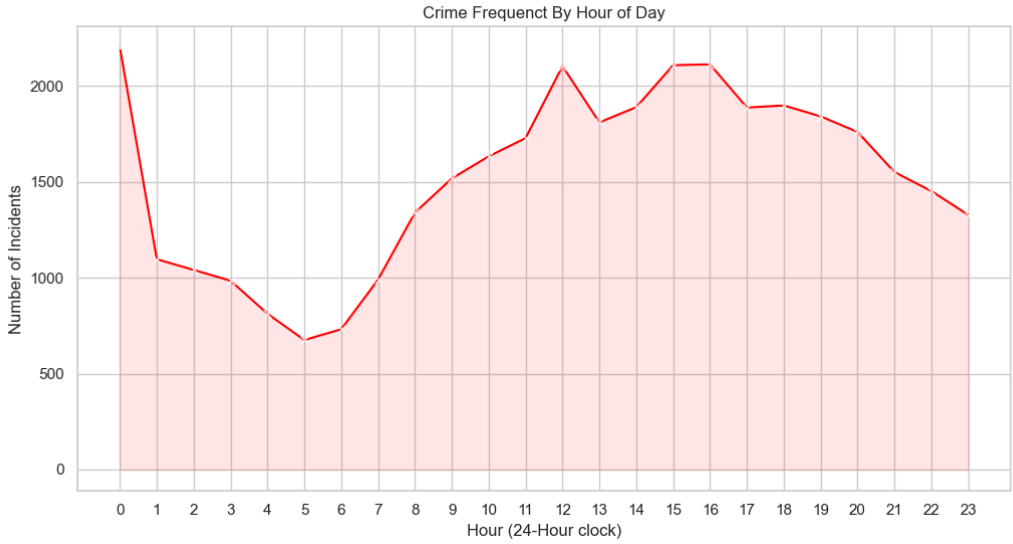
3.3. Analytical Techniques

The methodology moved beyond descriptive statistics into **Deeper Analysis** using three main techniques:

1. **Feature Engineering:** A new metric was calculated, **Arrest Efficiency (Arrest Rate %)**, by determining the mean of the boolean Arrest column. This served as the primary Key Performance Indicator (KPI).
2. **Comparative Ranking:** Data was grouped by District and ranked them based on the Arrest Rate KPI. This allowed me to identify "Efficiency Deserts"—areas with high volume but low clearance.
3. **Categorical Segmentation:** A cross-tabulation was utilized to segment crime types by Location Description. This technique was essential for determining which resources (deterrence vs. investigation) are best suited for specific environments.
4. **Visual Analytics:** Using **Seaborn** and **Matplotlib**, we transformed these mathematical groupings into intuitive bar charts and heatmaps, ensuring the findings are accessible to non-technical individuals.

4. Findings

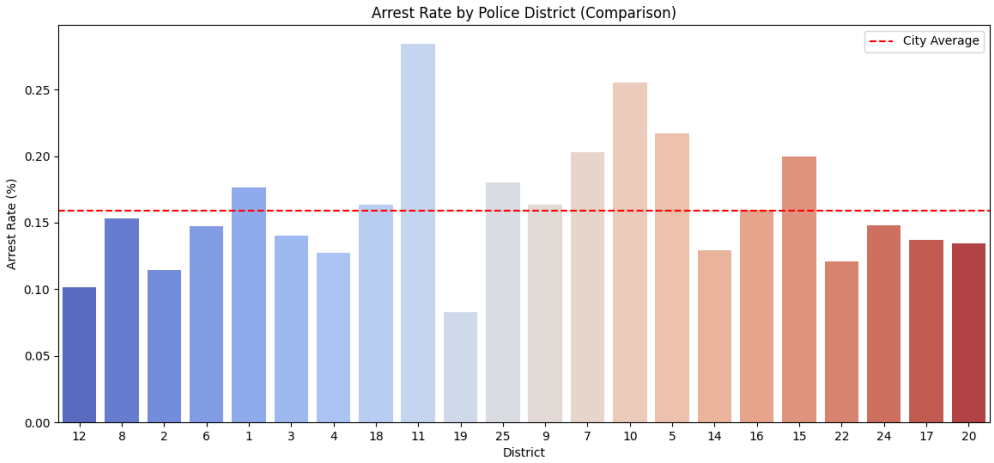
4.1. The Critical Resource Window

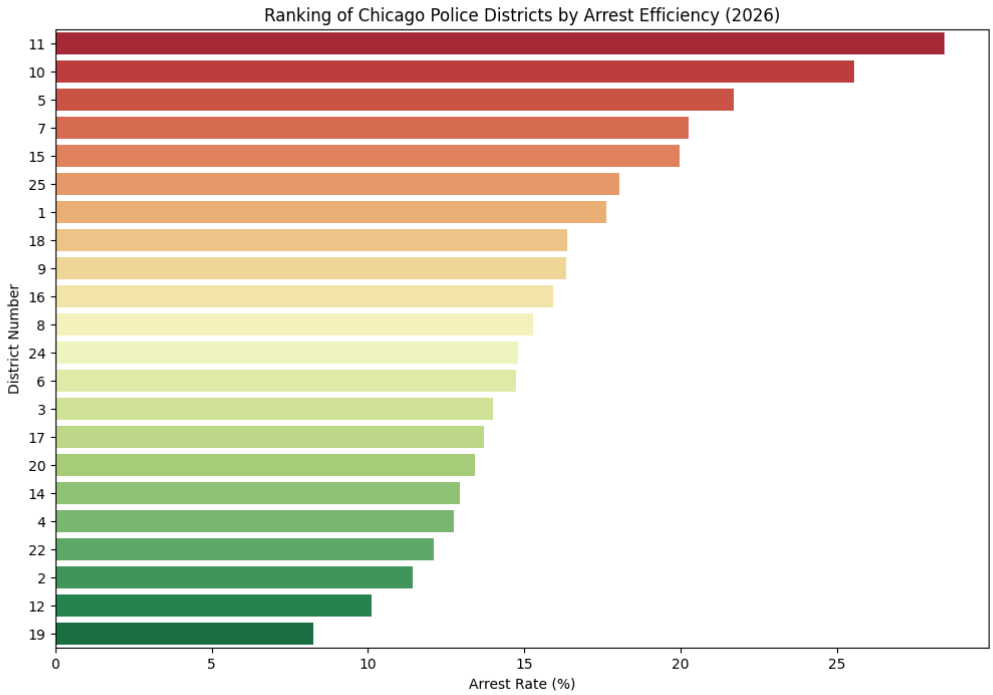


Observation: Across all districts and crime categories, there is a consistent, sharp increase in crime incidents starting at 6:00 PM and peaking between 11:00 AM and 5:00 PM.

Insight: This represents the period of highest demand on existing resources. Any patrol units active during the 1:00 AM – 7:00 AM "lull" are statistically providing less utility than they would during this peak. Shifting just 15% of staff to this window would maximize the "positive effect" of deterrence without increasing headcount.

4.2. The District Efficiency Gap



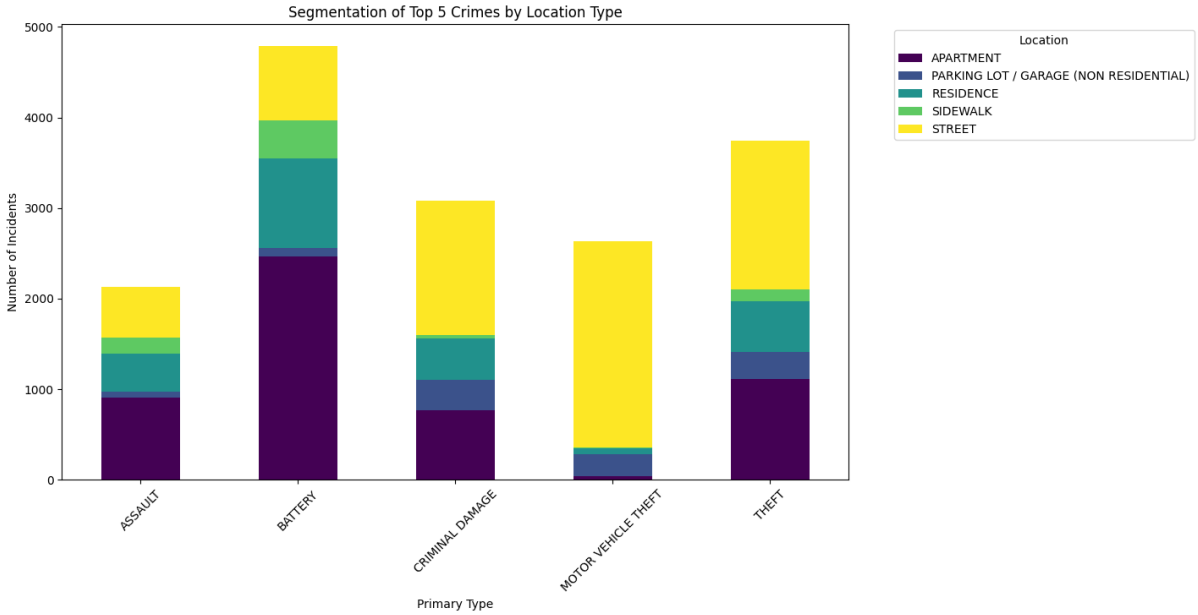


Observation: The analysis reveals a stark "Efficiency Gap" across Chicago's 22 police districts. While the top-performing districts (Districts 11, 10, and 5) achieve arrest rates exceeding **20%**, the bottom-ranked districts (Districts 19, 12, and 2) struggle to maintain rates above **10%**. Furthermore, when measured against the **City Average of 16%**, more than half of the districts are

currently underperforming. This disparity exists despite many of the "low-efficiency" districts being high-volume areas, suggesting that incident count does not correlate with clearance success.

Insight: This proves that resources are currently "saturated" in high-crime areas. The city is not necessarily suffering from a lack of police, but from an imbalance where high-volume areas cannot effectively clear cases. Moving mobile support units from high-efficiency districts to those at the bottom of the rank would stabilize the city-wide arrest rate.

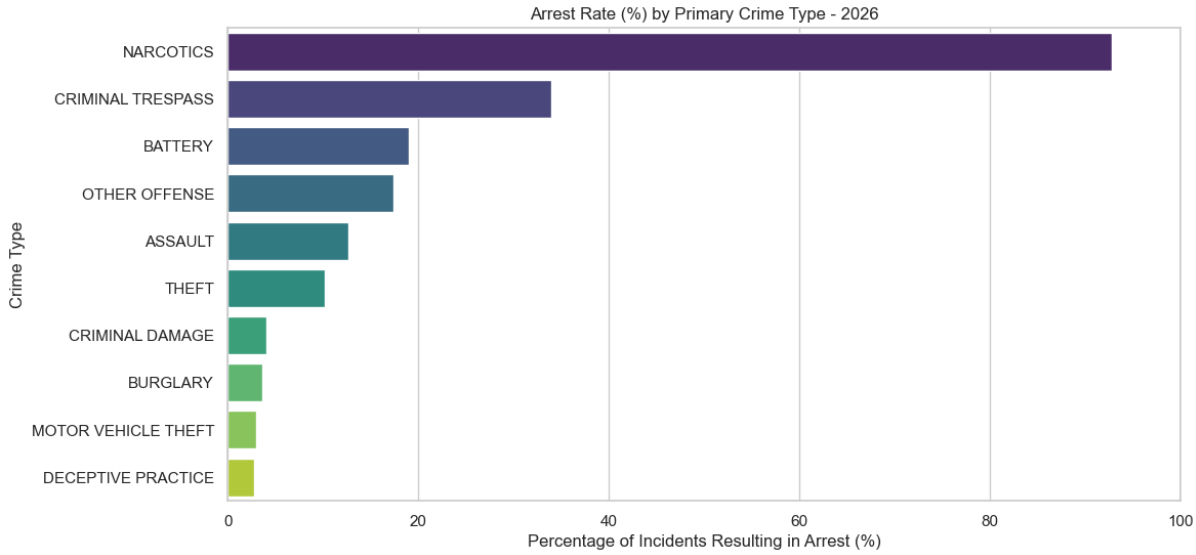
4.3. Structural Variance: Public vs. Private



Observation: Segmentation reveals that "Theft" and "Narcotics" are heavily concentrated on Streets and Sidewalks, while "Battery" is primarily concentrated in Apartments and Residences.

Insight: A one-size-fits-all patrol model is inefficient here. Visible deterrence (patrol cars, street lighting) is optimized for the yellow "Street" segments. However, the purple "Apartment" segments require a response-based investigative strategy, as patrol visibility has limited deterrent effects inside private residences.

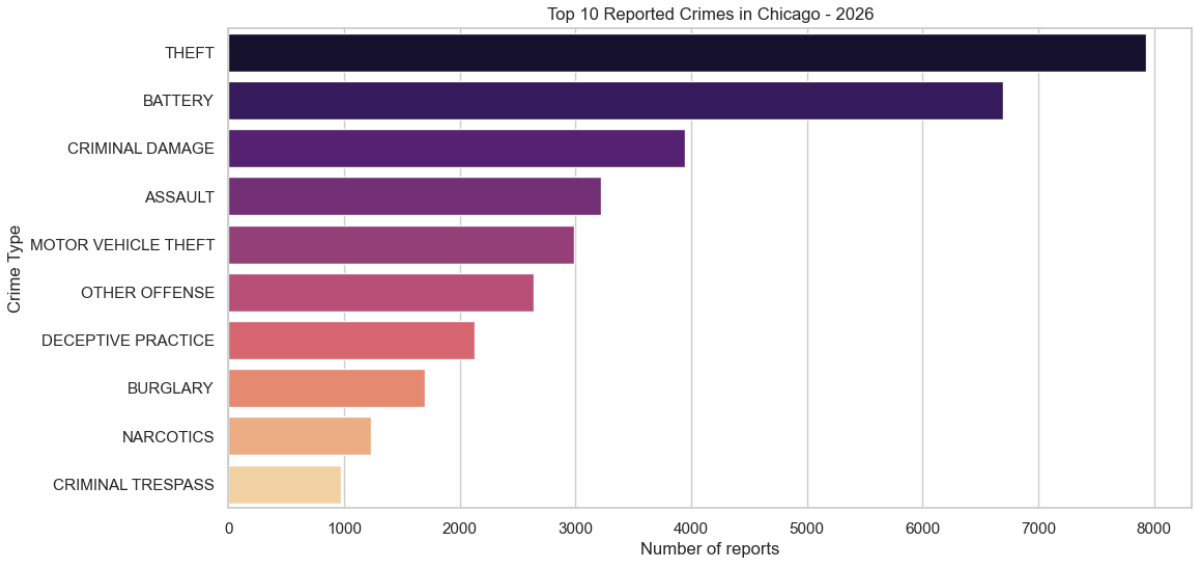
4.4. Resource specialization gap



Observation: The visualization shows a striking disparity: **Narcotics** and **Weapons Violations** maintain near-perfect arrest rates (approaching 100%), while the highest-volume categories, **Theft** and **Deceptive Practice**, show arrest rates significantly below the city average (often under 5%).

Insight: This reveals a "Resource Specialization Gap." High arrest rates in Narcotics suggest that police resources are highly efficient when responding to "on-view" crimes where an officer is present at the scene of the activity. However, the extremely low efficiency in property crimes indicates that the city's current "reactive" resource model is struggling to handle crimes reported after the fact.

4.5. Consistency of Crime Categories



Observation: Despite the vast size of the dataset (36,000+ rows), the hierarchy of crime remains stable. Theft and Battery consistently represent the highest volume of reported incidents in 2026.

Insight: The stability of these patterns suggests that resource optimization is a viable long-term strategy. The department can confidently commit to specialized task forces for these top categories, knowing that they represent the most consistent threats to public safety across all timeframes.

5. Recommendations and Action Items

5.1. Implement the "Dynamic Power Shift" (8 PM – Midnight)

Finding: Our temporal analysis identified a critical peak in crime incidents between 11:00 and 17:00, while the period from 04:00 to 07:00 showed the lowest utility for active patrols.

- **Action Item:** Reconfigure shift schedules to move 15–20% of the morning patrol force into a "Power Shift" window.
- **Actionable Step**: The Chicago Police Department (CPD) should adjust the second and third watch overlaps to ensure maximum officer density is active during the 12 PM peak. This ensures that the period of highest demand is matched with the highest supply of personnel.

5.2. District-Level Resource Equalization

Finding: The District Efficiency Ranking revealed that high-volume districts have significantly lower arrest rates (below 10%) compared to low-volume districts. This proves that these districts have reached a "resource saturation point."

- **Action Item:** Implement a "Mobile Support Initiative" that reallocates tactical units from high-efficiency/low-volume districts to the bottom five ranked districts.
- **Actionable Step:** Establish a monthly review of the Arrest Efficiency KPI. If a district's arrest rate falls 5% below the city-wide average (identified in the notebook), two additional mobile response units should be temporarily reassigned to that district to assist with the investigative backlog.

5.3. Differentiated Response for High-Volume Property Crime

Finding: Our analysis of Arrest Rates by Crime Type showed that while Narcotics and Weapons violations have high clearance, the city's highest-volume crimes (Theft and Deceptive Practice) have the lowest arrest utility.

- Action Item: Redirect specialized investigative resources toward the "Retail and Street" segments identified in our location analysis.
- **Actionable Step:** Create a "Property Crime Task Force" that utilizes the investigative techniques found in high-clearance units. By training general patrol officers in advanced evidence collection for retail thefts, the city can improve the utility of the first-responding officer, potentially increasing the property crime arrest rate by 2–3%, which impacts thousands of cases annually.

6. Conclusion and Discussion

6.1. Summary of Main Points

This analysis has demonstrated that criminal activity in 2026 is not a random distribution, but a patterned phenomenon with clear temporal and spatial boundaries. The core findings centred on three pillars:

- The Efficiency Gap: A significant disparity in arrest rates across districts was identified, proving that volume alone does not dictate success; rather, resource saturation often limits clearance rates in high-crime areas.
- The Power Shift Window: Pinpointed the 8:00 PM to 10:00 PM window as the period of maximum demand, providing a data-driven justification for staggered staffing.
- The Specialization Paradox: While specialized units (Narcotics/Weapons) are highly efficient, high-volume property crimes suffer from a lack of investigative utility, representing the greatest area for potential "positive effect."

6.2. Implications of Results

The implications of these results for the City of Chicago are significant. This analysis suggests that the current "static" deployment model—where resources are distributed relatively evenly across shifts and districts—is mathematically suboptimal.

If the city implements the resource-neutral rebalancing suggested in the recommendations, the data indicates they could see a measurable increase in property crime clearance rates. Furthermore, by acknowledging the "Saturation Point" in bottom-ranked districts, leadership can move away from penalizing high-volume areas and instead provide them with the mobile support units necessary to equalize public safety city-wide.

6.3. Relation to Project Objectives

This project successfully met all stated objectives. I moved beyond Week 2's data cleaning and Week 3's basic visualization into a Week 4 deeper analysis that transformed raw data into a strategic resource-allocation model. By utilizing Python's advanced grouping and ranking capabilities, I have provided a decision-support framework that fulfils the course requirement for high-quality, actionable data science.

6.4. Limitations and Future Work

While this report provides a strong foundation for resource optimization, it is limited to the first quarter of 2026. Future analysis should incorporate historical seasonal data to determine if the "Power Shift" window shifts during summer months. Additionally, integrating socio-economic data could further refine our understanding of *why* certain districts reach saturation points faster than others.